DGMS 2019 GEOLOGY • QUESTION 38

Silicate Mineralogy: Polymerization & Structural Classes

Topic: Mineralogy & Silicate Chemistry | Primary Reference: Dexter Perkins (Mineralogy, 3rd ed., Ch. 13 & 14)

Question Statement

Q.38. Match List-I (Mineral example) with List-II (Silicate class):

List-I (Mineral example):
A. Phlogopite
B. Enstatite
C. Anthophyllite
D. Beryl

List-II (Silicate class):
1. Inosilicate single chain
2. Cyclosilicate
3. Phyllosilicate
4. Inosilicate double chain

(a) A-3, B-1, C-4, D-2
(b) A-2, B-1, C-4, D-3
(c) A-2, B-4, C-1, D-3
(d) A-3, B-4, C-1, D-2

a **A-3, B-1, C-4, D-2 (Correct Answer)**

b A-2, B-1, C-4, D-3

c A-2, B-4, C-1, D-3

d A-3, B-4, C-1, D-2

✓ Correct Answer: (a) A-3, B-1, C-4, D-2 [Phlogopite = Phyllosilicate; Enstatite = Single-chain Inosilicate; Anthophyllite = Double-chain Inosilicate; Beryl = Cyclosilicate].

Textbook Evidence: Dexter Perkins (Chapter 13)

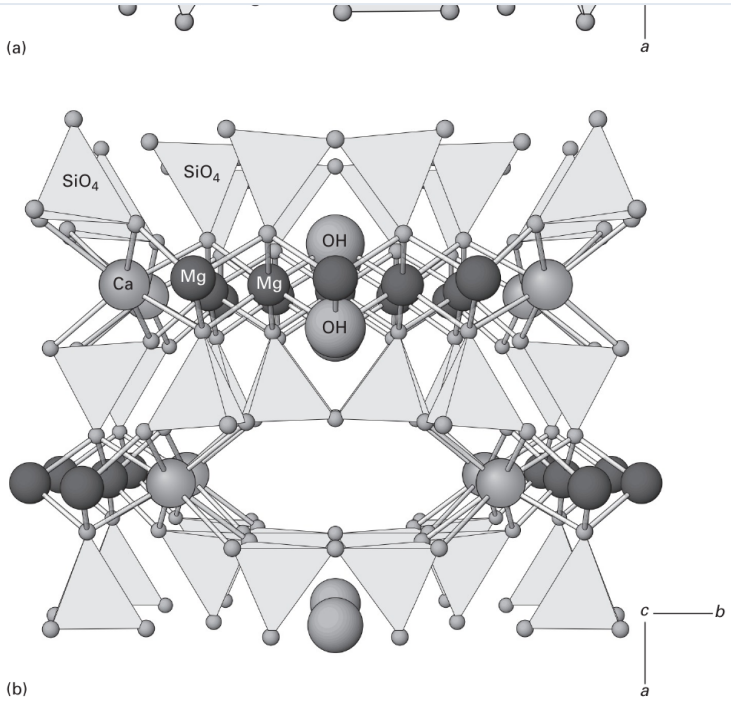


FIGURE 17 Atomic arrangement in (a) diopside (pyroxene) and (b) tremolite (amphibole). The tetrahedra are $(SiO_4)^{4-}$ with small spheres at the corners representing the oxygen. The larger spheres represent Ca^{2+} and Mg^{2+}, and $(OH)^-$ in tremolite. Both views are down the *c*-axis, so we are looking down the chains of tetrahedra.

Figure 13.17: Atomic Architectures of Single-Chain and Double-Chain Inosilicates (Dexter Perkins, *Mineralogy*, 3rd ed., p. 370). Down-c-axis projections illustrating: (a) Single chains of corner-sharing $[SiO_4]^{4-}$ tetrahedra linked by bridging oxygens (Si:O = 1:3, characteristic of pyroxenes like enstatite and diopside); (b) Double chains formed by cross-linking two single chains sharing alternating oxygens (Si:O = 4:11, $[Si_8O_{22}]^{12-}$, characteristic of amphiboles like anthophyllite and tremolite).

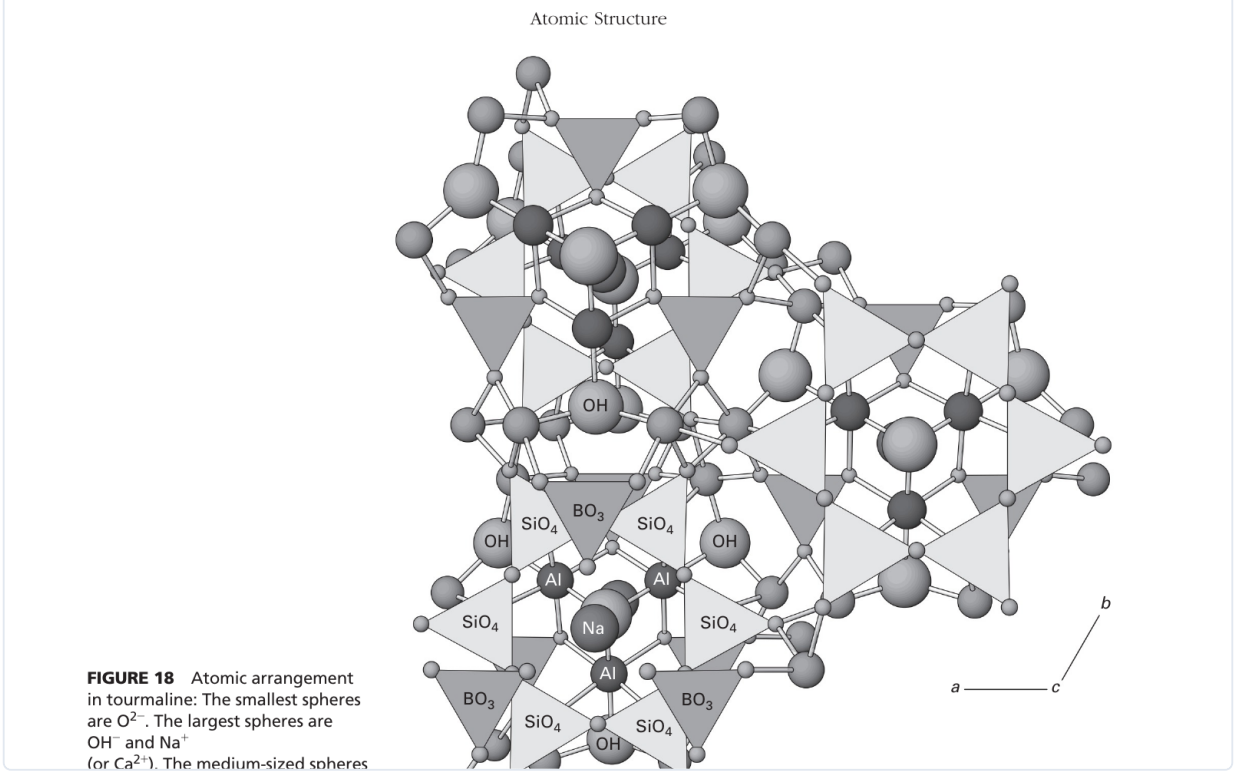


FIGURE 18 Atomic arrangement in tourmaline: The smallest spheres are O^{2-}. The largest spheres are OH^{-} and Na^{+} (or Ca^{2+}). The medium-sized spheres

Figure 13.18: Cyclosilicate Ring Geometry (Dexter Perkins, *Mineralogy*, 3rd ed., p. 371). The 6-membered ring $[Si_6O_{18}]^{12-}$ structure of cyclosilicates (beryl, $Be_3Al_2[Si_6O_{18}]$, and tourmaline). Each tetrahedron shares two oxygens with its ring neighbors, leaving large central channels parallel to the c-axis.

> *"Silicate minerals are classified primarily on the basis of how $[SiO_4]^{4-}$ tetrahedra are linked together:*
> • **Phyllosilicates (Sheet silicates):** *Three bridging oxygens shared per tetrahedron, forming continuous 2D sheets ($[Si_4O_{10}]^{4-}$, e.g., micas including phlogopite and biotite).*
> • **Inosilicates (Chain silicates):** *Single chains sharing 2 oxygens ($[Si_2O_6]^{4-}$, pyroxenes like enstatite); double chains sharing alternately 2 and 3 oxygens ($[Si_8O_{22}]^{12-}$, amphiboles like anthophyllite).*
> • **Cyclosilicates (Ring silicates):** *Tetrahedra share two oxygens to form closed 3-, 4-, or 6-membered rings ($[Si_6O_{18}]^{12-}$, e.g., beryl)."*
>
> **— Dexter Perkins, *Mineralogy*, Chapter 13 (Atomic Structure of Silicates), pp. 364–372**

Detailed Step-by-Step Mineral Matching

A. Phlogopite → 3. Phyllosilicate [MATCH 1]

Chemical Formula: $\text{KMg}_3[\text{AlSi}_3\text{O}_{10}](\text{OH})_2$

Phlogopite is the magnesium-rich endmember of the mica group. Its crystal structure consists of continuous two-dimensional sheets of corner-sharing silica-alumina tetrahedra (t-o-t sandwiches) linked by interlayer potassium cations. Because it forms flexible hexagonal sheets with perfect basal cleavage $\{001\}$, it belongs unconditionally to the **Phyllosilicates** (Sheet silicates).

B. Enstatite → 1. Inosilicate single chain **[MATCH 2]**

Chemical Formula: $\text{Mg}_2\text{Si}_2\text{O}_6$ (or MgSiO_3)
Enstatite is the magnesian endmember of the orthopyroxene group. Pyroxenes are characterized by single, unbranched chains of $[SiO_4]$ tetrahedra extending parallel to the c-crystallographic axis. Each tetrahedron shares exactly two oxygen atoms with its neighbors, giving a fundamental repeating ratio of $\text{Si}:\text{O} = 1:3$. Thus, enstatite is an **Inosilicate single chain** mineral.

C. Anthophyllite → 4. Inosilicate double chain **[MATCH 3]**

Chemical Formula: $\text{Mg}_7\text{Si}_8\text{O}_{22}(\text{OH})_2$
Anthophyllite is an orthorhombic amphibole commonly found in metamorphosed ultramafic and mafic rocks. Amphiboles are constructed of double chains of tetrahedra created by cross-linking two pyroxene-like single chains. The repeat unit is $[Si_8O_{22}]^{12-}$, yielding diagnostic ~124° and 56° prismatic cleavage intersections. Anthophyllite is an **Inosilicate double chain** mineral.

D. Beryl → 2. Cyclosilicate **[MATCH 4]**

Chemical Formula: $\text{Be}_3\text{Al}_2[\text{Si}_6\text{O}_{18}]$
Beryl (varieties include emerald and aquamarine) consists of rigid 6-membered rings of $[SiO_4]$ tetrahedra stacked along the c-axis, bound laterally by beryllium and aluminum cations. Rings formed of closed loops of tetrahedra define the **Cyclosilicates** (Ring silicates).

Master Classification of Silicate Minerals

Silicate Subclass	Structural Arrangement	Shared Oxygens per Tetrahedron	Si : O Ratio	Diagnostic Mineral Groups
Nesosilicates (Orthosilicates)	Isolated independent $[SiO_4]^{4-}$ tetrahedra	0	1 : 4 ($[SiO_4]$)	Olivine, Garnet, Zircon, Al_2SiO_5 (Kyanite, Sillimanite, Andalusite)
Sorosilicates (Disilicates)	Double isolated tetrahedra pairs ($[Si_2O_7]^{6-}$)	1	2 : 7 ($[Si_2O_7]$)	Epidote group, Zoisite, Hemimorphite
Cyclosilicates (Ring)	Closed rings of 3, 4, or 6 tetrahedra	2	1 : 3 ($[Si_6O_{18}]$)	**Beryl**, Tourmaline, Cordierite, Benitoite
Inosilicates (Single chain)	Continuous single chains	2	1 : 3 ($[Si_2O_6]$)	**Pyroxene group (Enstatite, Diopside, Augite)**, Pyroxenoids
Inosilicates (Double chain)	Continuous double chains	Alternating 2 and 3	4 : 11 ($[Si_8O_{22}]$)	**Amphibole group (Anthophyllite, Tremolite, Hornblende)**
Phyllosilicates (Sheet)	Continuous 2D planar sheets (t-o-t or t-o)	3	2 : 5 ($[Si_4O_{10}]$)	**Mica group (Phlogopite, Muscovite, Biotite)**, Clays, Chlorite, Talc
Tectosilicates (Framework)	Full 3D tetrahedral framework	4	1 : 2 ($[SiO_2]$)	Quartz, Feldspars (Plagioclase, Orthoclase), Feldspathoids, Zeolites

Key Takeaway & Exam Strategy

Rapid Silicate Matching Algorithm:

- **Mica (Phlogopite, Biotite, Muscovite):** Always **Phyllosilicate** (Si_4O_{10}).
- **Pyroxene (Enstatite, Diopside, Augite, Hypersthene):** Always **Single chain** (Si_2O_6).
- **Amphibole (Anthophyllite, Tremolite, Actinolite, Hornblende):** Always **Double chain** (Si_8O_{22}).
- **Beryl** / **Tourmaline:** Always **Cyclosilicate** (Ring silicate).
- This yields directly: **A-3, B-1, C-4, D-2** (Option a).